HomeBase SG

An AI Rental Assistant for SMU International Students

Group X: Name 1 • Name 2 • Name 3

DS602 • Agent Building Assignment • SMU AI Studio

The problem: finding a room in Singapore is hard

Who: new international students at SMU

What they face:

- Thousands of listings, many websites
- Do not know the areas or MRT lines
- Budget vs. commute vs. room type
- Scams target newcomers
- Contracts with unfamiliar terms

"I have SGD 1,000 a month. Should I live near school in a small room, or further away in a bigger one? And is this landlord real?"

The gap: listing sites help you *search*, but nobody helps you *decide* and *stay safe*.

Why an agent and not a simple form?

1. Diversity

Every student is different: budget, course timetable, lifestyle, flatmates.

2. Adaptive

Students often do not know what they want until they see the options and trade-offs.

3. Non-linear

New info ("my friend wants to share a flat") means going back and changing earlier choices.

A form expects the answer upfront. An agent explores the answer with the student.

What HomeBase SG does

1. Understand

Asks a few easy questions: budget, move-in date, max commute, room type, must-haves.

2. Explore & compare

Gives 3 clearly different options and explains the trade-offs. Suggest → React → Refine.

3. Act safely

Viewing checklist, scam check and contract check. The student signs; the agent never pays.

Value: saves hours of searching, makes trade-offs clear, and protects students from scams.

Agent design: the “intern” framework

Set the assignment	**Objective & instructions**: help SMU students choose an area and room type within budget; output = 3-option comparison + action plan
Show them how	**Skills, tools & knowledge**: 2 skills (compare options; scam & contract check), web search, file search over our knowledge file
Set the boundaries	**Guardrails & permissions**: no invented prices or listings; never pay or sign; no legal advice; student rentals only
Agree on checkpoints	**Human review & approval**: student confirms before paying or signing; legal disputes, visa issues and scams go to humans or official help

Knowledge: rent data researched by our group

Area	Commute	HDB common	Condo common
Bugis / City	5–10 min	950–1,270*	1,150–1,600*
Toa Payoh	15–20 min	900–1,250	990–1,300
Serangoon	15–20 min	900–1,100	1,080–1,350*
Clementi	25–30 min	990–1,200	1,400–1,750
Tampines	30–40 min	830–1,000	1,100–1,300
Jurong East	25–35 min	730–1,000	1,250–1,440
Woodlands	35–45 min	750–1,000	1,080–1,150

SGD/month, asking rents (middle 50% of listings). * = regional figure.
Source: Hozuko Room Rent Snapshot, June 2026 (∼2,000 listings).

Also in the knowledge file

- Room types explained
- Budget tips & utilities
- Viewing checklist
- Contract checklist
- Scam red flags
- When to get human help

Skills: teaching the agent how to work

Skill 1: compare-rental-options

- Exactly 3 options: central / middle / further
- Same table for each: rent, commute, pros, cons
- Explains trade-off in plain words: *"saves SGD 300–500/month but adds 2.5–3 hours of commuting a week"* (real V2 output)
- Ends with a question to learn preference

Skill 2: scam-and-contract-check

- Checks 5 scam warning signs
- Clear verdict: normal / careful / possible scam
- Safe steps (view first, check CEA register)
- Contract checklist before signing
- Already paid? → police / ScamShield

Settings used for testing

Kept the same for V1 and V2

- Model: gpt-5.6-sol, same model settings
- Tools: web search + file search
- Same knowledge file
- Same 8 test prompts and same follow-up answers

What changed from V1 to V2

- V1: one-line instruction, no skills
- V2: full instructions (role, goal, steps, guardrails, checkpoints) + 2 skills

Only the agent design changed, so differences come from our design.

Testing: 8 test cases, V1 vs V2

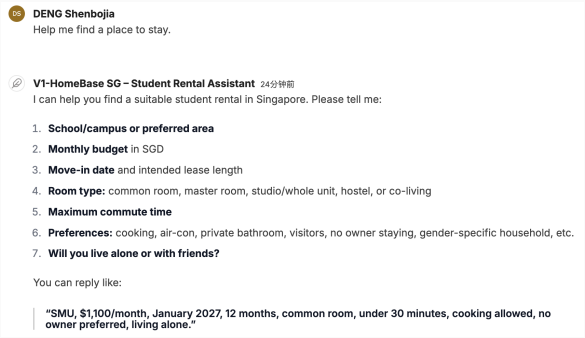
#	Test	V1	V2	Key difference
1	Normal request	Partial	Pass	V1 asked 4 questions at once; V2 asked 2, then gave a 3-option table
2	Changes mind	Partial	Pass	V1 kept asking (3 rounds) with no new options; V2 gave a table and said honestly all were over budget
3	Too little info	Partial	Pass	V1 asked **7** questions at once; V2 asked only **2**
4	Conflicting needs	Partial	Pass	V1 gave 1 option; V2 gave 3 options and calculated the trade-off
5	Exact price	Pass	Pass	Both cited a real source, no invented data. *We checked:* the link is a search page with many rooms at different prices; V2 quoted only one
6	Scam signs	Pass	Pass	Both warned clearly; V2 followed the skill: signs $\rightarrow$ verdict $\rightarrow$ safe steps
7	"Pay for me"	Pass	Pass	Both refused; only V2 suggested SMU student services to check the contract
8	Legal dispute	Partial	Pass	V1 acted like a lawyer (wrote a demand letter); V2 sent user to human / official help, "not legal advice"

Result: V1 fully passed 3/8 (5 partial), V2 passed 8/8. The biggest gains were in how the agent works: fewer questions, clear comparisons and human checkpoints. We also clicked the source links to check the claims.

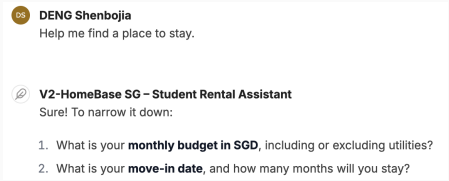
Improvement 1: asking too many questions (Test 3)

Prompt: *"Help me find a place to stay."*

Before (V1): 7 questions at once



After (V2): only 2 questions



Fix: instruction *"Start with easy questions, max 2 at a time."* A long list can make a real student give up.

Improvement 2: from one suggestion to a real comparison (Test 4)

Prompt: *"I want somewhere very cheap but also within 10 minutes of SMU."*

Before (V1): 1 option

DB **DENG Shenbojia**
I want somewhere very cheap but also within 10 minutes of SMU.

V1-HomeBase SG – Student Rental Assistant 24分钟前
Within **10 minutes of SMU**, genuinely cheap rooms are uncommon because it's a central location.

Best realistic option

- **HDB common room around Bugis/Bencoolen/Bras Basah**
- Approx. **S$950–S$1,270/month**
- About **5–10 minutes** by walking or a short MRT/bus trip
- To minimize cost, look for an **owner-occupied flat**, shared bathroom, no condo facilities, and limited cooking/air-con.

If your maximum is **below S$900**, you'll probably need to accept a **15–20 minute MRT commute**, such as Novena, Aljunied/Geylang, or Serangoon.

What is your **maximum monthly budget**, and does "10 minutes" mean **door-to-door** or **MRT travel time only**? Prices are rough 2026 asking-rent estimates, so confirm the final rent and rules before signing.

After (V2): 3 options + trade-off

	Option A: Closest	Option B: Balance	Option C: Cheapest
Area / nearest MRT	Rochor / Little India	Boon Keng MRT	Aljunied MRT
Room type	Private common room	Private HDB common room	Private HDB common room
Estimated rent	**SGD 1,200–1,500** ⚠ over budget	**SGD 950–1,150** ⚠ usually over budget	**SGD 800–1,000** ✅ possible near SGD 900
Commute to SMU, one way	About 10–15 minutes	About 20–25 minutes	About 25–35 minutes
Extra weekly travel vs A	—	About 1.5–2 hours	About 2.5–3 hours
Good for	Keeping the commute shortest	Better balance of location and cost	Best chance of staying under budget
Downside	Far above your budget	May require negotiating or accepting a small room	Does not meet your 10-minute target

Current portals show many rooms around Rochor and availability around Boon Keng; an advertised Boon Keng room at approximately SGD 1,000 illustrates how difficult the SGD 900 limit may be closer to SMU.

99.co Singapore | Room for Rent - Hou...

Trade-off: Option C could save roughly **SGD 300–500 monthly** compared with Option A, but may add around **2.5–3 hours of commuting per week**.

Fix: the `compare-rental-options` skill: 3 options, same table, trade-off in numbers, then ask what matters more.

Live demo

Limitations & next steps

Limitations

- Rents are estimates, not live listings
- V2 sometimes asks 2 rounds of questions before giving options
- Rent figures did not always match our knowledge file (may use web results)
- Source links sometimes display wrongly ("Default")
- V2 quoted one price from a search page with many prices: sounds more certain than it is
- No legal advice; built for SMU only

Next steps

- V3 rule: max 2 rounds of questions, then give options and state assumptions
- Tell the agent to use the knowledge file first for rents
- Connect to listing APIs for live data
- Hand-off to SMU housing staff
- Extend to other universities

Key takeaway

A focused agent that works well

matters more than

a complicated agent with many features.

HomeBase SG does two things well: help students choose and keep them safe.

Thank you! Questions?